Code Execution Note

Next line of the code written after i/p stmt only after you give i/p & hit enter (any code unexpected i/p & o/p) will execute.

In max code:

Java code

View code

```java
int max = Integer.MIN_VALUE;
for(int i = 0; i < arr.length; i++) {
    if(arr[i] > max) max = arr[i];
}
return max;
```

Edge case:

- `Arrays.toString(arr);`
- `int[] arr = null; arr = null`
- `int[] arr = new int[0]; // arr exists, length = 0, not null`

Java code
View code

```java
new int[][] {
    {1,2,3},
    {4,5,6}
}
```

-> this is also correct here we have initialized 2D array

- While directly returning 2D array you initialize it first using new keyword without declaring it. `return new int[]{i,j};`

Math.log:

- `Math.log(a);` -> is always of type double
- So while using it/storing it to int typecast it to int
- `int b = (int)Math.log(a);`

To get no. of digits:

`T.C = O(1)` -> a) it calculates no. of digits in 1 step

- `(int) Math.log10(a) + 1; // to get no. of digits in decimal`
- `(int)(Math.log(a)/Math.log(2)) + 1` -> to get no. of digits in binary for given decimal no.
- `(int)Math.log10(arr[i]) + 1; // to get no. of digits in decimal`
- `String s = String.valueOf(arr[i]); int digits = s.length();`

```
int count = 0;
if (n == 0) return 1; // edge case

while (n > 0) {
    n /= 10;
    count++;
}
return count;
```

Method	Output	Use Case
`str.toCharArray()`	`{'G','a','u','r','i'}`	When you need to iterate/modify chars
`Arrays.toString(str.toCharArray())`	`[G, a, u, r, i]`	When you need to print/debug